

## Linear Algebra

## Appetizer

1. **(August 2017 Problem 2)** Let  $K$  be a field, and let  $A$  be an  $n \times n$  matrix over  $K$ . Suppose that  $f(x) \in K[x]$  is an irreducible polynomial such that  $f(A) = 0$ . Show that  $\deg(f) \mid n$ .
2. Give two examples of matrices in Jordan form with minimal polynomial  $x(x-1)^2(x-2)$  and characteristic polynomial  $x^2(x-1)^4(x-2)$ .

## Entrée

3. Let  $V$  be a finite-dimensional vector space over a field. Let  $A : V \rightarrow V$  be an endomorphism, and suppose  $W \subseteq V$  is an invariant subspace (so that  $A(W) \subseteq W$ ). Let  $\bar{A} : V/W \rightarrow V/W$  be defined by  $\bar{A}(v + W) = Av + W$  for all  $v \in V$ .
- (a) Show that  $\bar{A}$  is well-defined.
- (b) Show that if  $\bar{A}$  is not invertible, then  $A$  is not invertible.
4. **(January 2020 Problem 4)** Let  $V$  be a finite-dimensional vector space over a field. Let  $A : V \rightarrow V$  be a linear transformation.
- (a) Show that there exists  $B : V \rightarrow V$  such that  $ABA = A$ .
- (b) Show that there exists  $C : V \rightarrow V$  such that  $ACA = A$  and  $CAC = C$ .
5. **(January 2015 Problem 4)** Let  $F$  be a field, and let  $V$  be a finite-dimensional vector space over  $F$  that has dimension  $n \geq 1$ . Let  $q$  be a nonzero element of  $F$  such that  $q^i \neq 1$  for  $1 \leq i \leq n$ . Let  $X : V \rightarrow V$  and  $Y : V \rightarrow V$  be two  $F$ -linear transformations such that  $XY = qYX$ . Show that  $XY$  is nilpotent.
6. Let  $V, W$  be two finite-dimensional vector spaces over  $\mathbb{C}$ . Let  $A : V \rightarrow V$  and  $B : W \rightarrow W$  be linear transformations. List the eigenvalues of  $A \otimes B : V \otimes W \rightarrow V \otimes W$  in terms of the eigenvalues of  $A$  and  $B$ .
7. **(January 1994 Problem 4)** Let  $X$  be a subspace of  $M_n(\mathbb{C})$ , the  $\mathbb{C}$ -vector space of all  $n \times n$  complex matrices. Assume that every nonzero matrix in  $X$  is invertible. Prove that  $\dim_{\mathbb{C}}(X) \leq 1$ .

**8. (January 2017 Problem 2)** Let  $n > 0$  be an integer. Let  $F$  be a field of characteristic zero,  $V$  a vector space over  $F$  of dimension  $n$ , and  $T : V \rightarrow V$  an invertible  $F$ -linear map such that  $T^{-1} = T$ . Denote by  $W$  the vector space of linear transformations from  $V$  to  $V$  that commute with  $T$ . Find a formula for  $\dim(W)$  in terms of  $n$  and the trace of  $T$ .

**9. (January 2019 Problem 5)** Let  $F$  be a field and  $n$  be a positive integer. Fix an  $n \times n$  matrix  $S$  over  $F$  this is invertible and symmetric. Writing  $A^t$  for the transpose of a matrix, we let

$$V := \{n \times n \text{ matrices } A \text{ over } F \mid A^t = SAS^{-1}\}.$$

Note that  $V$  is a vector space (you do not need to prove this). Find the dimension of  $V$  in terms of  $n$ .